



EDU TERIA

72nd BPSC

Prelims Online Batch

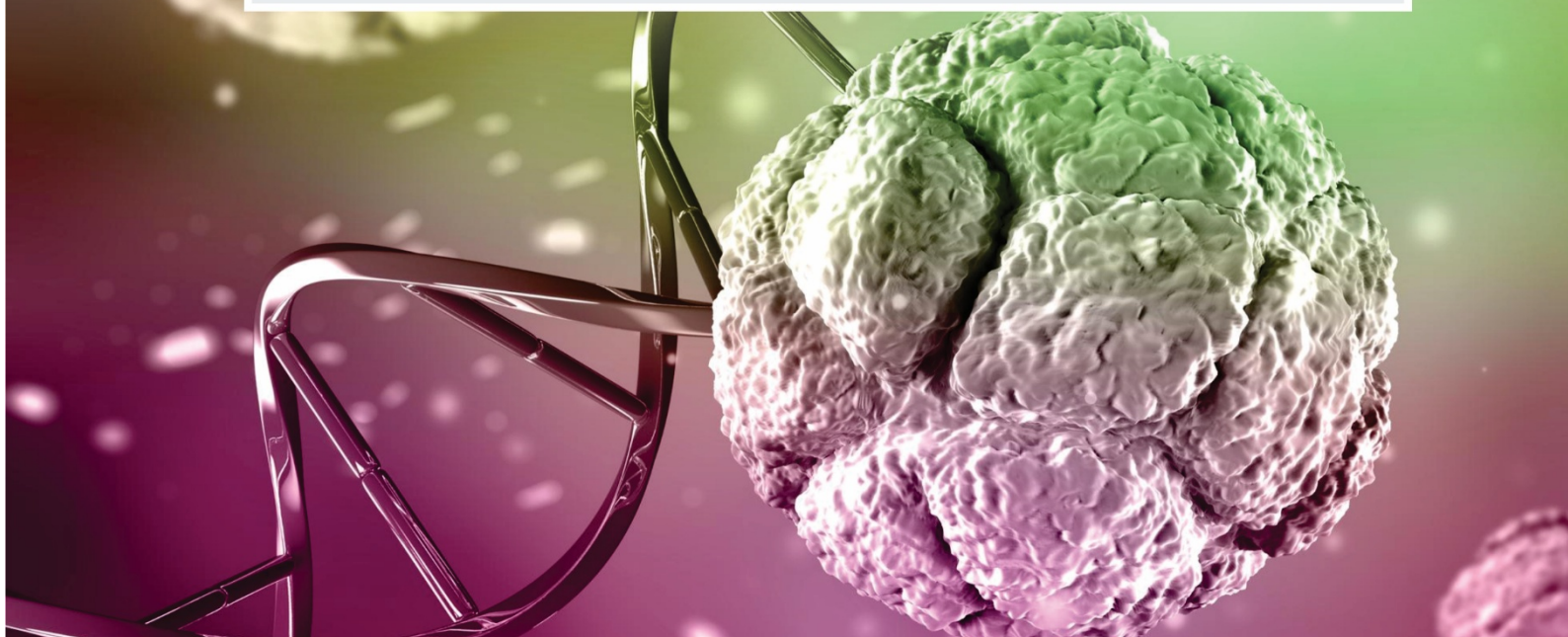
BIOLOGY

Topic wise
Notes

LECTURE

03

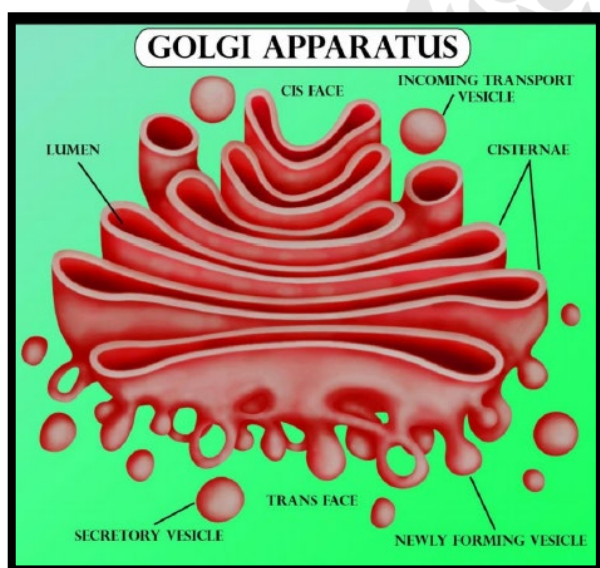
Cell Organelles (Part-II)



CELL ORGANELLES (Part-II)

GOLGI APPARATUS

The Golgi apparatus (or Golgi complex/body) is a membrane-bound organelle in eukaryotic cells, acting as the cell's "post office." It receives proteins and lipids from the Endoplasmic Reticulum (ER), modifies, sorts, and packages them into vesicles for secretion or delivery to other organelles. It consists of stacked, flattened sacs called cisternae.



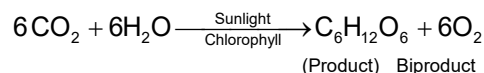
PLASTID

- Discovered by Ernest Haeckel
- Observed by Schimper
- Only present in plant Cell
- Three types of plastids are present in plant Cell—
- Chloroplast also known "Kitchen of the Cell"
- Chloroplast also known as Semi-Autonomous Cell organelle.
- Chloroplast contain photosynthetic pigment known as chlorophyll which provides green colour to chloroplast.
- Chloroplast is the cite for photosynthesis.

Photosynthesis :—

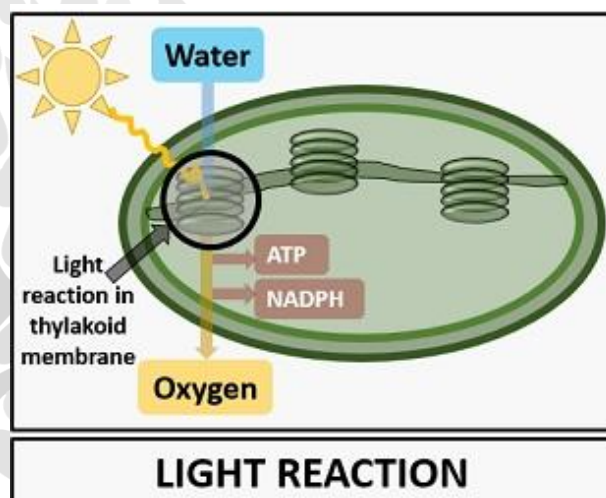
The process of formation of Glucose (food) in the presence of sunlight and chlorophyll with the

help of sunlight and chlorophyll with help of carbon dioxide and water.

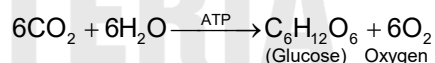


Photosynthesis includes two steps—

- Light Reaction—** Occurs in grana. In this reaction sunlight stored in the form of ATP molecules.

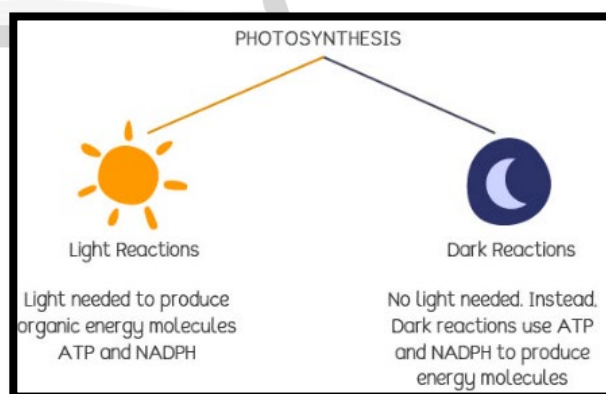


- Dark Reaction—** Does not need direct sunlight. Dark Reaction occurs in stroma.



Dark Reaction is also known as Calvin Cycle.

Photosynthesis— Sunlight / Solar Energy— Chemical Energy.

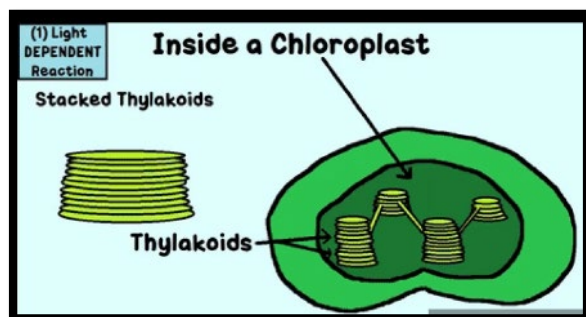


**PLASTID**

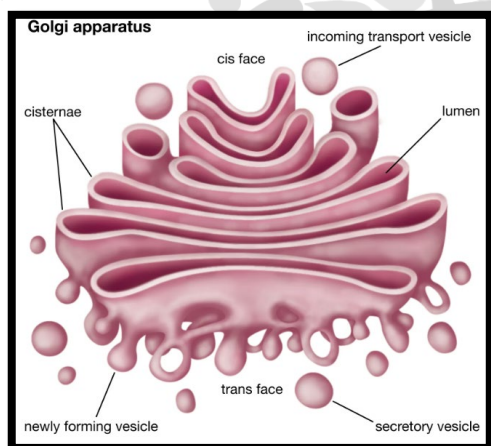
Plastids are large, double membrane-bound cell organelles found mainly in plant cells and algae.

They are responsible for:

- Manufacturing food
- Storing food materials
- Synthesizing pigments
- Giving colour to plant parts

**Golgi body :**

The Golgi apparatus (or Golgi complex/body) is a vital membrane-bound, single-membrane eukaryotic organelle first observed by Camillo Golgi (1898),

**Structure—**

Comprises flattened, disk-shaped sacs called cisternae

Cis-face—Convex, located near the nucleus/endoplasmic reticulum, receives materials.

Trans-face—Concave, located near the plasma membrane, dispatches processed materials. The cis and trans faces are interconnected.

Function—

Packaging & Transport— Principal function is packaging materials (Proteins/lipids) into vesicles to be delivered to intracellular targets or secreted outside.

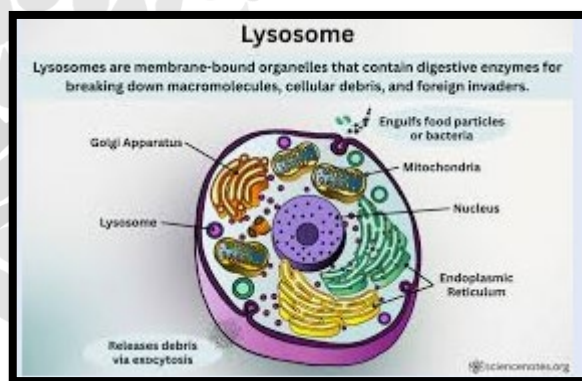
Organelle Formation—Essential for the formation of lysosomes, acrosome of sperm, and cell plates during plant cell division.

Lysosome—

Lysosomes are single-membrane bound vesicular organelles, derived from the Golgi apparatus, acting as the "Suicide Bags" of eukaryotic animal cells.

Contains more than 50 types of digestive enzymes of hydrolytic in nature.

pH=5



TERIA

